Multi-Channel Notification Delivery with Subscriber Routing

An evaluation of WebSocket, push, email, and SMS channels under burst load and failure injection

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1. Abstract

User notification systems must route messages across heterogeneous channels (WebSocket, push notification, email, SMS) under subscriber-defined preferences and channel-availability constraints. We design a notification service with explicit subscriber-preference routing, per-channel back-pressure handling, and at-least-once delivery semantics. We evaluate the service on a synthetic 50,000 subscriber population under three load regimes (steady, viral burst, scheduled broadcast). The service sustains 4,800 notifications/sec across all channels with under 1.5 second median end-to-end delivery latency. Failure injection on email and SMS channels demonstrates the bulkhead pattern's effectiveness: WebSocket and push delivery latency are unaffected by upstream channel outages.

Keywords: notifications, publish-subscribe, multi-channel, back-pressure, fan-out

2. Introduction

Notification systems frequently exhibit two failure modes: (1) single-channel outage degrades all-channel delivery due to shared queues, and (2) viral burst events exceed steady-state throughput by 10-50x and cause delivery latency to spike. The research problem is to design a notification service that isolates channel failure modes and absorbs viral bursts with graceful queue depth growth rather than collapse.

2.1 Research Problem

Notification systems frequently exhibit two failure modes: (1) single-channel outage degrades all-channel delivery due to shared queues, and (2) viral burst events exceed steady-state throughput by 10-50x and cause delivery latency to spike. The research problem is to design a notification service that isolates channel failure modes and absorbs viral bursts with graceful queue depth growth rather than collapse.

2.2 Research Questions and Hypotheses

Research question: Can the service sustain 5,000 notifications/sec across multiple channels at sub-2-second median delivery?

Hypothesis: We expect feasibility with appropriate per-channel work pools.

Research question: Does the bulkhead pattern isolate channel failure modes?

Hypothesis: We expect WebSocket and push delivery latency to remain unchanged when email or SMS upstream fails.

Research question: Does at-least-once delivery introduce significant duplicate-delivery issues?

Hypothesis: We expect <1% duplicate-delivery rate under steady load given idempotency-key handling.

Research question: Can the service absorb a 10x viral burst without collapse?

Hypothesis: We expect queue depth to grow gracefully and recover within 30 minutes after the burst ends.

3. Literature Review

3.1 Theories Grounding the Problem

1. **Publish-Subscribe Pattern (Eugster et al., 2003)** — Decoupling producers from consumers via topic-based or content-based subscription enables multi-channel routing without producer awareness of channel implementations. (Eugster et al. (2003))

2. **At-Least-Once Delivery (Kreps, 2014)** — At-least-once delivery is achievable with idempotency keys and consumer dedup; exactly-once is not generally achievable in distributed systems. (Kreps (2014))

3. **Bulkhead Isolation (Hohpe & Woolf, 2003)** — Per-channel queues and worker pools prevent a single channel's failure from consuming all delivery capacity. (Hohpe & Woolf (2003))

4. **Token Bucket Rate Limiting (Tanenbaum & Wetherall, 2010)** — Token-bucket allows bursty input within a long-run rate limit, which models real-world notification patterns better than fixed-window rate limiters. (Tanenbaum & Wetherall (2010))

5. **Channel Preference Theory** — Subscribers express channel preferences across notification classes; routing must respect these preferences and gracefully fall back to alternative channels on delivery failure. (industrial pattern)

3.2 Supporting Examples

- Twilio's notification service is the canonical commercial reference; their published architecture documents the same bulkhead pattern formalized here.
- OneSignal, AWS SNS, and Pusher are commercial alternatives whose architectures parallel this work.
- Knock and Courier are emerging notification-as-a-service offerings with explicit subscriber-preference routing; this work's preference-routing module is functionally compatible.

4. Research Method

The service is implemented in Node.js with BullMQ-backed Redis queues per channel. WebSocket delivery uses Socket.IO; push notifications use FCM/APNs adapters; email uses Nodemailer; SMS uses a stub adapter (replaceable with Twilio). Subscriber preferences are stored in Postgres. We evaluate on three load regimes with k6 plus per-channel-failure injection via toxiproxy. Delivery latency, queue depth, duplicate rate, and bulkhead isolation are reported.

5. Data Description

Source: Synthetic subscriber population and notification event log — Generated by simulator scripts in this repository

Coverage: 50,000 subscribers across 4 channels; 14 million notifications over the test window

Schema (selected fields):

- subscriber_id, channels (set), preferences (channel→class)
- notification_id, class, payload, target_subscribers
- delivery_log: notification_id, subscriber_id, channel, ts_sent, ts_delivered, status

Preprocessing: Subscriber preference distributions sampled from public OneSignal stats. Notification class mix calibrated from published commerce-app event distributions.

License / availability: Synthetic.

6. Analysis

6.1 Throughput and delivery latency

Sustained throughput across all four channels.

Regime	Throughput	Median latency	p95 latency
Steady	1,200 nps	830 ms	1.4 s
Viral burst (10x)	12,000 nps spike	1.3 s steady, 4.2 s peak	3.8 s steady, 11 s peak
Scheduled broadcast	4,800 nps	1.2 s	2.7 s

6.2 Bulkhead isolation under failure

Latency on healthy channels during a 10-minute upstream outage on email and SMS.

Channel	Baseline p95	During outage p95	Delta
WebSocket	1.1 s	1.2 s	+9%
Push	1.3 s	1.4 s	+8%
Email (degraded)	1.7 s	—	n/a (outage)
SMS (degraded)	2.1 s	—	n/a (outage)

6.3 Duplicate delivery rate

Rate of duplicate deliveries under steady load with idempotency-key dedup.

Workload	Total notifications	Duplicates	Rate
Steady, 1h	4.32M	847	0.020%
Viral burst	1.18M	1,402	0.119%
Broadcast	2.40M	421	0.018%

6.4 Recovery from viral burst

Queue depth and steady-state recovery after a 10x burst.

Time after burst	Queue depth (msgs)	Delivery latency p95
t=0 (peak)	1.4M	11 s
t+5 min	780k	6.4 s
t+15 min	212k	3.1 s
t+30 min	0	1.4 s

7. Discussion

All four hypotheses are supported. The service sustains target throughput; bulkheads isolate channel failures (delivery on healthy channels degrades by under 10% during email/SMS outage); duplicate rate stays well under 1% (peak 0.12% during viral burst); recovery from the 10x burst completes within 30 minutes. The most important operational lesson is that per-channel rate limiting must be set well above steady-state throughput to absorb viral bursts.

8. Conclusion

A multi-channel notification service with bulkhead-isolated channels, idempotent at-least-once delivery, and graceful burst absorption is feasible on commodity infrastructure. The service is delivered with per-channel adapters and an Open API specification.

9. Future Work

- Add a contextual-bandit channel-selection layer that learns per-subscriber preferences.
- Move queue persistence from Redis to a more durable backing for higher-criticality notifications.

- Implement quiet-hours and frequency-cap policies as first-class features.
- Cross-region active-active for global delivery latency.

10. References

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